Gazebo Behavior Tree Autonomy Validation

Technical evidence report | ROS 2 Humble | Gazebo Classic 11 | Docker | A* + pure pursuit (not Nav2)

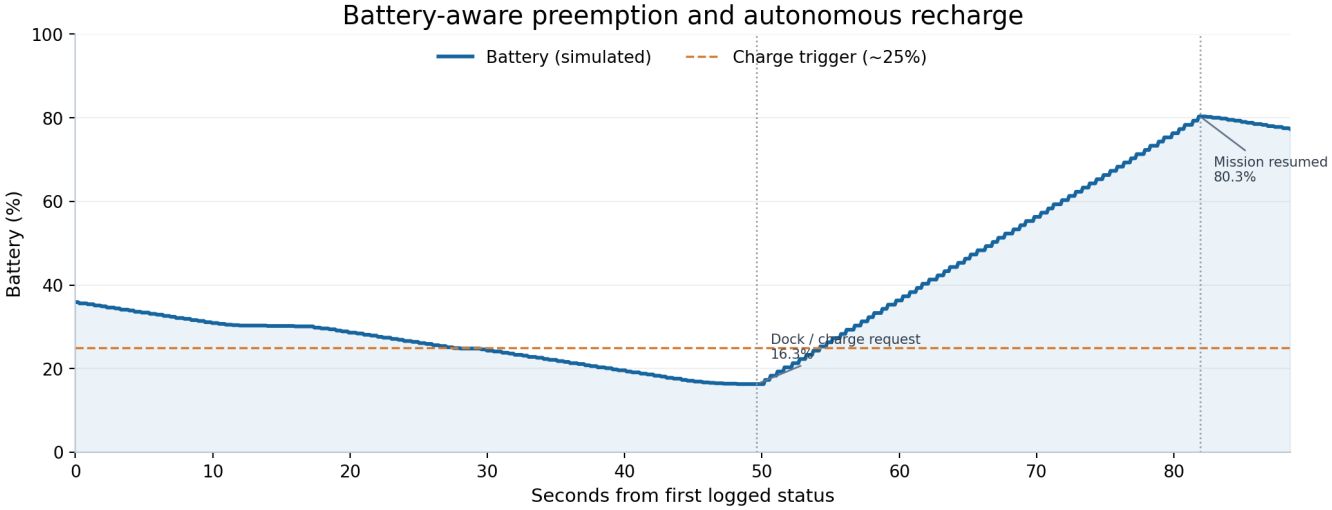
22 / 22	7 / 7	52.27 s	880
Core unit tests	Offline waypoint pairs	Independent inspection goal	BT status records

Executive summary

In a recorded Gazebo Behavior Tree run, patrol_perimeter and inspect_east_room appear in the completed mission list. During the captured delivery segment, low battery caused the executive to interrupt deliver_package at step 2, navigate autonomously to the dock, initiate charging, and resume the same step after recharge.

Evidence boundary: The captured stream ends while delivery step 2 is still running; the final DONE state and successful delivery completion are not documented. All battery percentages are from a simulation model.

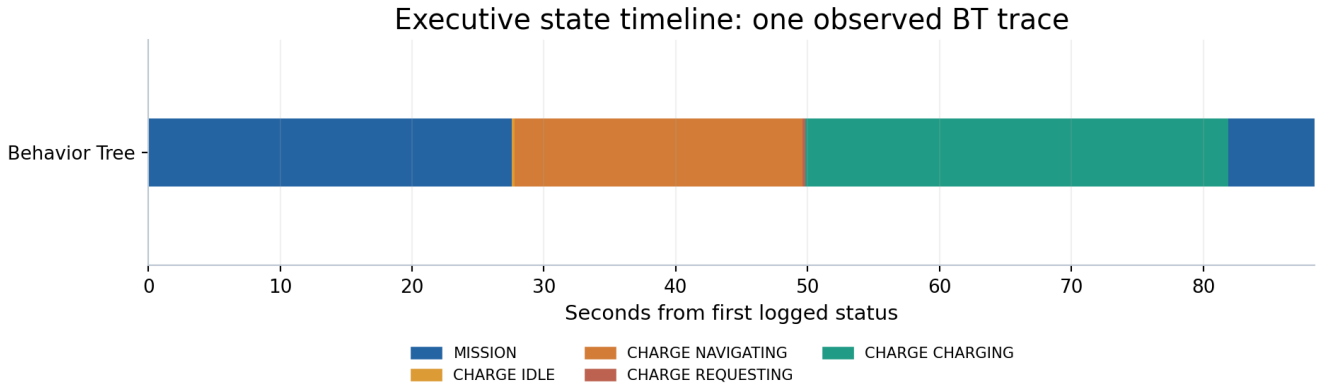
Battery during observed interruption

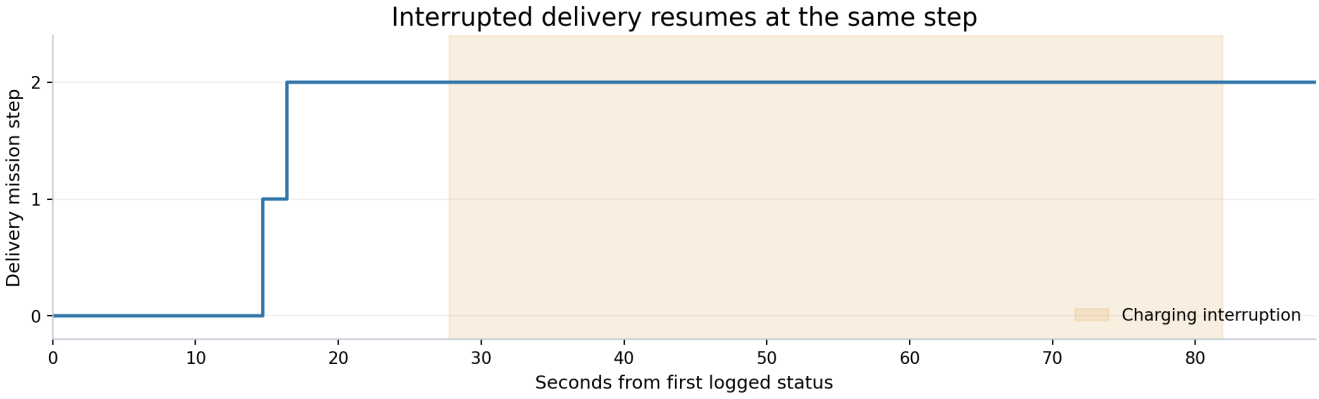


Recorded values: 35.9% at the beginning of the captured delivery segment; 24.9% at return-to-dock navigation; 16.3% at the charge request; 80.3% at return to MISSION; 77.3% at the end of capture. Total observation window: 88.47 s.

Observed transitions and resume behavior

This timeline is computed directly from the JSON status payloads, without interpolating mission outcomes.





Milestone timeline

Elapsed	Executive state	Battery	Delivery step
0.0 s	MISSION	35.9%	0
16.4 s	MISSION	30.1%	2
27.6 s	CHARGE_IDLE	24.9%	2
27.8 s	CHARGE_NAVIGATING	24.9%	2
49.6 s	CHARGE_REQUESTING	16.3%	2
50.2 s	CHARGE_CHARGING (charging)	17.3%	2
81.9 s	MISSION	80.3%	2
82.4 s	MISSION	80.2%	2

Mission completion was established from the executive status payload, not from visual inspection of Gazebo. The status recorded failed=[] throughout this capture.

Method, traceability and limits

Simulation and navigation configuration

ROS 2 Humble, Gazebo Classic 11, differential-drive robot with 2D LiDAR and odometry. Navigation comprises an in-project A* grid planner, path smoothing, a pure-pursuit controller and bounded scan-block replanning. It is explicitly not Nav2.

The observed navigation improvement followed an experimental planner-only inflation adjustment: base radius 0.30 m + added clearance margin 0.20 m = 0.50 m. Stopping clearance was not intentionally reduced. This single observation does not establish optimal parameters or repeated-run reliability.

Independent verification (different runs)

Validation	Recorded result
Build / unit tests	8 ROS 2 packages finished; bip_core: 22 tests, 0 failures
Offline map connectivity	7 of 7 waypoint-pair paths found; not proof of execution safety
Direct inspection navigation	GoToPose (4.5, 3.0, 0.0): success=true; message=reached; 52.265911 s

Data supplied with this report

The evidence package includes bt_status.jsonl (880 unaltered parsed ROS JSON messages), bt_status.csv, summary.json, this PDF, a Markdown report, three plotted PNGs and a reproducible plotting script at tools/plot_bt_evidence.py. The action result is preserved in direct_goal_result.txt.

What is NOT demonstrated

No final successful three-mission DONE event in this log; no repeated successful trials, confidence intervals, collision metrics or head-to-head BT/FSM/Utility results; no physical-robot test; no recorded video included. Avoid presenting the battery model as a measured real-battery performance result.

Next validation and publication steps

1. Record the complete BT final state in the next run. 2. Capture the authentic Gazebo GUI with a live state terminal in OBS. 3. Repeat comparable runs and collect BT/FSM/Utility benchmark statistics. 4. Commit source patches and evidence together on a review branch, then merge when reproducible.

Repository:
github.com/HARSHAVARDHAN-SEKAR/Autonomous-Mobile-Robot-Decision-Behavior-Intelligence-Platform-AMR-BIP-